

1. Input & Timing

glfwPollEvents()
deltaTime, FPS Window
Continuous Movement
Collision & Portals

2. Scene & Physics

Scene::update()
Kinematic Cycles
Day/Night State Machine
Update 16 Lights

3. Shadow Depth Pass

2048x2048 FBO
glCullFace(GL_FRONT)
shadow_depth.vert/.frag
Ortho Light Matrix

4. Main 3D Shading

Default FBO (1280x720)
16 Active Dynamic Lights
16-Sample PCF Filter
basic.vert / basic.frag

5. HUD & UI Overlay

Orthographic 2D Pass
Alpha Blending
Consoles Font Atlas
Minimal Telemetry

6. Export & Swap

Screenshot Capture
glReadPixels() to PNG/BMP
glfwSwapBuffers()
Frame Completion